

Introduction to Software Engineering

Software engineering is the systematic application of engineering approaches to the development of software. This course covers the fundamental principles and practices of software engineering.

Design Patterns

Design patterns are reusable solutions to commonly occurring problems in software design. They represent best practices for solving common design problems.

- Creational patterns: Factory Method, Abstract Factory, Builder, Prototype, Singleton
- Structural patterns: Adapter, Bridge, Composite, Decorator, Facade, Proxy
- Behavioral patterns: Chain of Responsibility, Command, Iterator, Observer, Strategy

Testing Strategies

Testing is essential for ensuring software quality. The testing pyramid suggests:

- Unit tests: Fast, isolated tests for individual functions
- Integration tests: Verify interactions between components
- End-to-end tests: Validate complete user workflows

Agile Methodologies

Agile development emphasizes iterative delivery, team collaboration, and continuous improvement. Key frameworks include Scrum, Kanban, and XP.